Phase 1 · Machine Learning Evaluation · Project 26

Card Fraud Triage via Gaussian Mixture Density Scoring

Team T033, Section A · Rishihood University

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Why card fraud is hard

Extreme imbalance

Only 492 of 284,807 transactions are fraud (0.17%). Predicting “legit” always gives 99.83% accuracy and catches nothing.

Delayed labels

Fraud labels arrive days or weeks later, so we cannot rely on supervised training.

Drift

Fraud tactics and normal spending patterns keep changing over time.

Our approach

Train only on legitimate transactions, learn what normal looks like, and rank every transaction by how much it deviates. Fraud labels are used only to tune and evaluate.

Dataset and chronological split

ULB credit card data: 284,807 transactions over 48 hours, 31 numeric columns, no missing values. 1,081 duplicates are kept to mirror real telemetry.

Split	Rows	Frauds	Base rate
Train (60%)	170,884	360	0.21%
Validation (20%)	56,961	57	0.10%
Test (20%)	56,962	75	0.13%

Zero-leakage rules

Time-ordered split, scalers and models fit on legitimate training rows only, thresholds chosen on validation and frozen for test.

What the data audit showed

Not Gaussian

Shapiro-Wilk rejects normality for all 28 V-features.

Heavy tails

Excess kurtosis: V2 = 115, V7 = 77, V8 = 262. Only 9 of 28 features have |kurtosis| below 3.

Strong signals

KS separation: V14 0.85, V10 0.83, V12 0.80, V17 0.79, V11 0.79.

Skewed amount

Legit mean $88, median $22, max $25,691, so we use ln(1 + Amount).

Implication: one Gaussian underestimates tail mass, so a mixture ($K \geq 3$) is needed.

Feature engineering

Feature set	What changed	Val PR-AUC
raw	V1–V28 + raw Amount + Time	0.0432
base	ln(1 + Amount) + sin/cos of time	0.0466
+ norms	Add $\|V\|_2$, 3-sigma outlier count, max $\|V\|$	0.0593
engineered	Also prune V22, V23, V25 (KS < 0.05)	0.0693

Final input: 31 features, +48.8% over base. RobustScaler (0.0436) and Yeo-Johnson (0.0451) were worse than the log transform. Scores shown for a K = 3 diagonal GMM.

Why norms help

Fraud often shows several coordinates far from normal at once. The geometric norm captures that.

GMM theory and model selection

The model

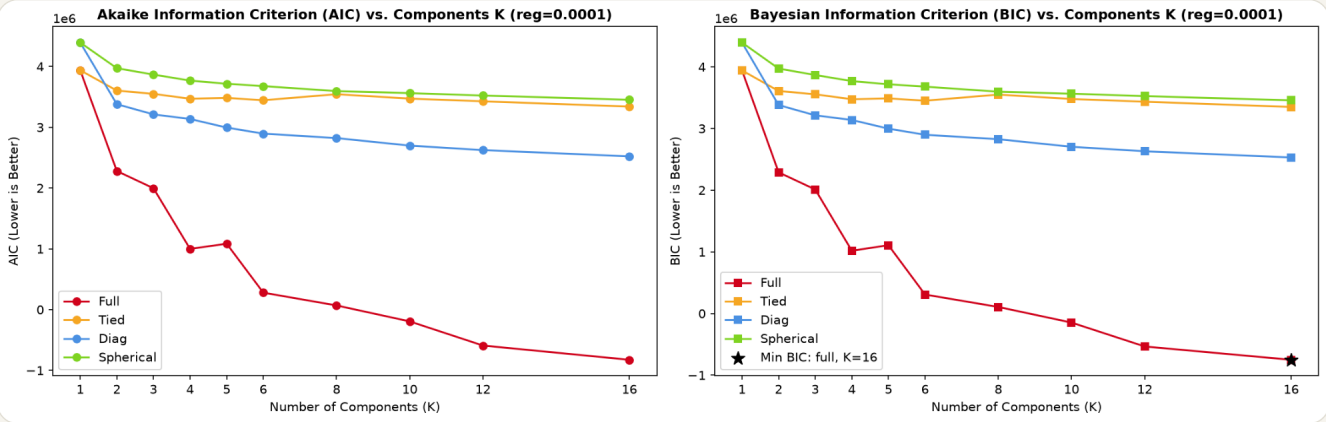
$p(x) = \Sigma\ \pi_k\ N(x \mid \mu_k, \Sigma_k)$. Anomaly score = $-\log p(x)$.

EM algorithm

E-step: responsibilities. M-step: update weights, means, covariances.

AIC vs BIC

Both trade likelihood against parameter count; BIC penalizes more.



Likelihood ≠ triage

Min BIC picks full, K=16, reg 1e-6: val PR-AUC 0.2123. Best validation PR-AUC picks full, K=12, reg 1e-2: 0.2684. 120 configs searched.

Final model refit on all 170,524 legit training rows.

GMM results and instability

0.4142

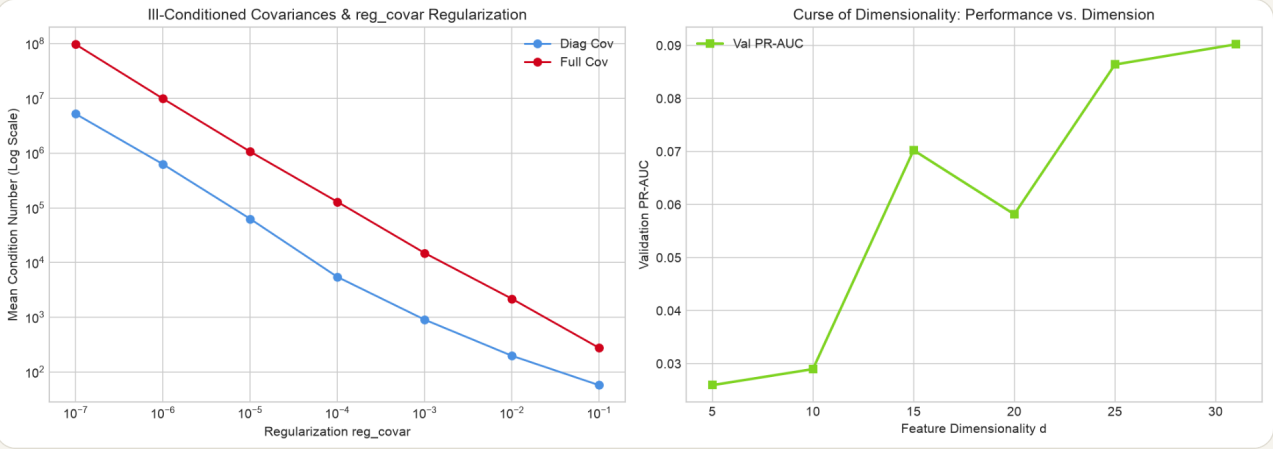
Test PR-AUC, 314× the base rate of 0.0013

69.3%

Recall at 0.5% false-positive rate

41.3%

Recall at 0.1% false-positive rate



What we measured

Condition number falls from ~6×10^5 to ~2×10^2 as reg rises 1e-6 → 1e-2. Seed spread: 0.087 ± 0.003 over 5 seeds. PR-AUC rises with dimension (0.026 at d=5, 0.090 at d=31) but dips at d=20. Validation 0.195 vs test 0.414: fraud shifts over time.

Stability studies use a K=5 diagonal GMM; the final model is full, K=12.

Four classical baselines

PCA reconstruction

Project onto top-k components of legit data. Score = reconstruction error. k chosen on validation.

Local Outlier Factor

Compares a point’s local density to its neighbors. Novelty mode, 40k legit rows.

One-Class SVM

RBF kernel boundary around normal data. Grid over ν and γ, 40k legit rows.

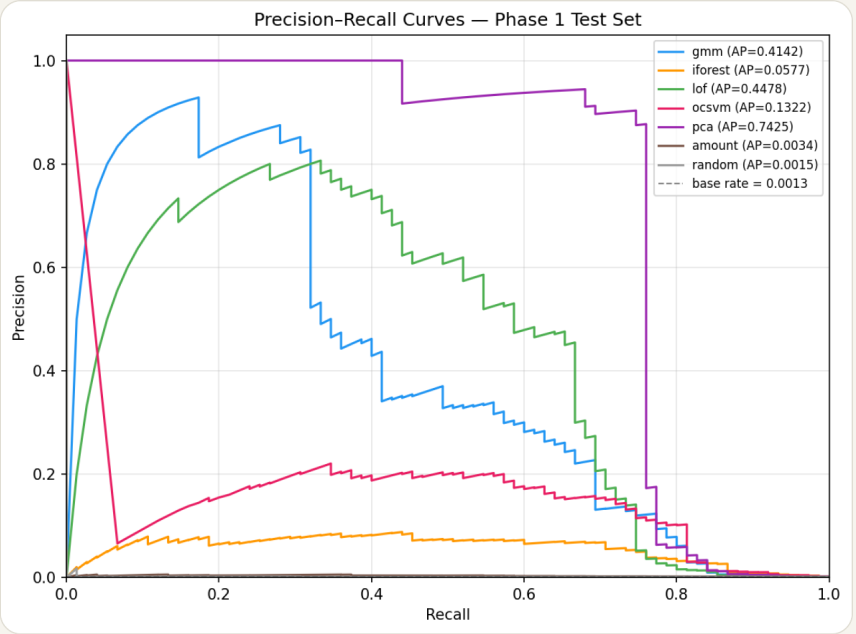
Isolation Forest

Random splits isolate outliers in fewer steps. Short path = anomalous.

All trained on legitimate data only; hyperparameters picked by validation PR-AUC. Random and lAmountl scores serve as sanity-check references.

Test-set comparison

Model	PR-AUC	ROC-AUC	Rec @ 0.5% FPR
PCA reconstruction	0.7425	0.930	77.3%
LOF	0.4478	0.905	72.0%
GMM	0.4142	0.927	69.3%
One-Class SVM	0.1322	0.960	69.3%
Isolation Forest	0.0577	0.957	34.7%
Amount · Random	0.0034 · 0.0015	0.71 · 0.48	1.3%



PCA wins (95% CI [0.64, 0.84], significantly above all others). ROC-AUC misleads: Isolation Forest has 0.957 ROC-AUC but 0.058 PR-AUC. With 75 test frauds, LOF vs GMM is not significant.

Triage trade-offs, limits and Phase 2

PCA policy	Alerts	Frauds caught	Precision	Recall
FPR ≤ 0.1%	141	57 of 75	40.4%	76.0%
FPR ≤ 0.5%	390	58 of 75	14.9%	77.3%
FPR ≤ 1.0%	750	58 of 75	7.7%	77.3%

Limits

Only 75 test frauds, 48-hour window, one dataset, anonymized features, assumed cost ratios.

Phase 2 next

Undercomplete or denoising autoencoder on the same split and metrics, compared against GMM and PCA.